Samrudh G Rao

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SUMMARY:

AI/ML engineer in Bengaluru who builds web platforms with LLM and OCR features. Three of the products I built run in production on Google Cloud. Day to day I work in React, Node.js and Python, with Firebase and Firestore for data, and I check AI output by hand before it ships. Earlier internships covered enterprise apps, workflow automation and ERP work.

EDUCATION

B.E. Computer Science & Engineering CGPA: **7.7** 2022 – June 2026
Jyothy Institute of Technology, VTU

Pre-University (PCMB) – Sadhana PU College (BASE Integrated) 2020 – 2022

TECHNICAL SKILLS

Languages	Python, JavaScript, SQL, C
Frameworks	React.js, Node.js, Express.js, FastAPI, Flask, Next.js, Flutter, React Native (Expo), OpenCV, PyTorch, YOLOv5
AI / ML	LLM Integration, NLP, CNNs, Face Recognition, Automated Reporting, Groq API, OpenAI API, OCR (Google Cloud Vision), Computer Vision, AI Output Validation
Databases	Firestore, MongoDB, MySQL, MS SQL Server
Cloud & Tools	Firebase, Google Cloud Run, AWS, Docker, GitHub, Power Apps, Power Automate, Google Drive API, Capacitor, Figma, Postman
Testing & QA	Manual, Functional, Regression and Usability Testing, Test Case Execution, Defect Tracking, Documentation

PROFESSIONAL EXPERIENCE

SARN Technologies Pvt. Ltd. | AI/ML Engineer, Bengaluru — May 2026 – Present

Tech: React 18, Node.js, Python Flask, Groq LLaMA, Firestore, Firebase Hosting, Google Cloud Run, JWT

- Built the platform sarntech.in runs on: three document workflows covering 13,000+ records, plus HR (attendance, leave, payroll) for 40+ staff. It is split across 3 Cloud Run services.
- Wrote the SDS Scanner in Flask with Groq LLaMA. It pulls fields and GHS pictograms out of PDFs. Added a chatbot that answers questions about an uploaded PDF.
- Cached the busiest Firestore reads with TTLs, which took read volume from millions a day down to a sustainable level.
- Role-based access on 200+ endpoints. Finished sheets are archived to Google Drive on a schedule.
- Payslips print onto the company letterhead, leave accrues monthly, and check-in is geofenced.
- Read AI outputs by hand and re-ran regression checks after every integration change.
- Also built and support two client products, Sustain Ledger Partners (GST invoicing SaaS) and Airavik (practice management for CA firms); details under Projects.

Wirtgen Group (A John Deere Company) | Software Development & Digitalization Intern, Pune — Jan 2026 – May 2026

Tech: React.js, Node.js, Firebase, MS SQL Server, Power Apps, Power Automate, SharePoint

- **QR Warehouse Management System:** wrote the frontend and connected it to the backend for QR-based visitor check-in and check-out. Data sits in MS SQL Server. Tested it before it went live.
- **Telemarketing Feedback Management System:** Power Apps front end, Power Automate for approvals and notifications, SharePoint for the data.
- **JDOC Operations & CRM Support:** monitored machines and dashboards, checked data, and handled user accounts and machine migrations.
- Tested new modules and fixed defects before release.

SARN Technologies Pvt. Ltd. | Full-Stack Developer Intern, Bengaluru — Jul 2025 – Dec 2025

Tech: React.js, Node.js, Firebase

- Set up the first frontend and backend structure of the workflow automation platform (React, Node.js, Firebase).
- Wrote core modules and the data-integration features, then tested and debugged them before release.

Vanillakart | Web Developer Intern, Bengaluru (Remote) — May 2024 – Nov 2025

Tech: HTML, CSS, JavaScript, Bootstrap, jQuery

- Built responsive pages that matched Vanillakart's branding, using Bootstrap and jQuery.
- Worked on page speed, SEO and the content management setup, and joined code reviews and debugging sessions.

PROJECTS

1. **Sustain Ledger Partners** - sustainledger.in *(live in production)* | Full-Stack Developer (solo, client product)
FastAPI, React 18, Vite, Capacitor (Android), Firestore, Google Drive API, Cloud Run
- Invoicing SaaS for Indian businesses. Each company gets its own data, branding, users and Google Drive storage. Produces GST PDF invoices and Excel exports, sends them by email or WhatsApp, and ships as a native Android app. 56 REST endpoints.

2. **SARN Technologies Platform** - sarntech.in *(live in production)* | AI/ML Engineer
React 18, Node.js, Express, Firestore, Cloud Run, Groq LLM
- Operations platform for an SDS compliance company. Document workflows, an HR suite and AI-assisted parsing of chemical data sheets.

3. **Airavik** - airavik.web.app *(live in production)* | Developer (client product)
React 18, Vite, Node.js, Express, Firestore, Firebase Auth, Cloud Run
- Practice-management platform for CA and audit firms. Tracks GST, TDS and PT filings through stages, reconciles GSTR-2B against purchase registers, runs HR and payroll, and keeps a tamper-evident audit log.

4. **WRV Energy: IoT Monitoring Platform and Website** - wrvenergy.com *(website live)* | Developer, Hosting & Marketing
FastAPI, Next.js 14, Expo, Google Cloud Vision, Groq (Llama 3.3), Firebase Hosting
- Reads utility bills and equipment displays with OCR, then an LLM turns the text into fields. A person reviews every value before it is saved. I also built and hosted the company website, set up its SEO, and shipped all of it in one month.

5. **SARN Workflow Automation System** | Full-Stack Developer
React, Node.js, Firebase, Firestore, AI APIs
- Built enterprise workflow automation platform using React, Node.js, Firebase and AI APIs.
- Implemented role-based access control, Google Sheets data import, and operational modules.
- Integrated AI-powered summarization and automated report generation.

6. **Scheduloop ERP (Deployed at JIT)** | Backend & AI Developer
Flutter, Node.js, Firebase, AI APIs
- Developed ERP backend with approval workflows, geo-tagged attendance, and venue booking.
- Implemented face recognition and LLM-powered event report generation.
- Deployed on JIT college infrastructure.

CERTIFICATIONS

- AWS Academy Graduate - Cloud Developing (2025)
- Career Essentials in Generative AI - Microsoft (May 2024)
- Introduction to Artificial Intelligence - LinkedIn Learning
- Agile Project Management - LinkedIn Learning

LEADERSHIP & ACTIVITIES

- Club Services Director - Rotaract Club of JIT
- Core Member - Team Aakarshana (Media & Photography Club)
- NCC Army Wing - B & C Certificate Holder (Oct 2022 – Oct 2025)

ADDITIONAL PROJECTS

Face Mask Detection (YOLOv5), AI Student Chatbot, Movie Recommendation System, Twitter Sentiment Analysis, AI Game Logic System.